

Linear algebra and geometry 1 - TTPU

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Grade 5.85 out of 30.00 (20%)

Question 1

Incorrect

Mark -15.00 out of
100.00

In the euclidean space let us consider the points

$$A = (1 + a, 2, 0), \quad B = (2, 1 + a, 0), \quad C = (1, 1, 2), \quad D = (1, 1, 0).$$

Which of the following statements is true?

- ☒ (a) There exists exactly two $a \in \mathbb{R}$ such that the points are collinear. ✗
- ☐ (b) There exists exactly two $a \in \mathbb{R}$ such that the points are coplanar.
- ☐ (c) For every $a \in \mathbb{R}$ the points are coplanar.
- ☐ (d) For every $a \in \mathbb{R}$ the points are collinear.

Risposta errata.

The correct answer is: There exists exactly two $a \in \mathbb{R}$ such that the points are coplanar.

Question 2

Incorrect

Mark -15.00 out of
100.00

Given the plane $\pi : 3x - 5y + z = 0$ and the straight line

$$r : \begin{cases} 2x + y - z = 1 \\ 3x + 2y + z = 1. \end{cases}, \text{ find the correct statement.}$$

-
- ☒ (a) None of the other statements is correct. ✖
- ☐ (b) $r \cap \pi = \emptyset$.
- ☐ (c) $r \subseteq \pi$.
- ☐ (d) r and π are orthogonal.

Risposta errata.

The correct answer is: r and π are orthogonal.

Question 3

Incorrect

Mark -15.00 out of
100.00

If the linear system $AX = B$ with 5 equations and 5 variables has ∞^m solutions with $m \leq 3$, then necessarily

-
- ☒ (a) $\text{rank}(A) \leq 2$. ✖
- ☐ (b) $\text{rank}(A) \geq 2$.
- ☐ (c) $\text{rank}(A|B) = 2$.
- ☐ (d) $\text{rank}(A) = 2$.

Risposta errata.

The correct answer is: $\text{rank}(A) \geq 2$.

Question 4

Correct

Mark 100.00 out of
100.00

Let $\cdot$ denote the usual matrix multiplication. If

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 9 \\ 0 & 0 \end{bmatrix},$$

then

- ☒ (a) $\det(\mathbf{A} \cdot \mathbf{B}) = 9$. ✓
- ☐ (b) $\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{B}) \cdot \det(\mathbf{A})$.
- ☐ (c) $\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$.
- ☐ (d) $\det(\mathbf{A} \cdot \mathbf{B}) = 0$.

Risposta corretta.

The correct answer is: $\det(\mathbf{A} \cdot \mathbf{B}) = 9$.

Question 5

Incorrect

Mark -15.00 out of
100.00

Let $f : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be an endomorphism with characteristic polynomial $P(t) = (t^2 - 2)(t^2 - t)$.

Which one of the following statements is true?

- ☒ (a) $\dim(\text{Im}(f)) = 2$. ✗
- ☐ (b) $\dim(\text{Ker}(f)) = 1$.
- ☐ (c) $\dim(\text{Ker}(f)) = 2$.
- ☐ (d) $\dim(\text{Im}(f)) = 1$.

Risposta errata.

The correct answer is: $\dim(\text{Ker}(f)) = 1$.

Question 6

Incorrect

Mark -15.00 out of
100.00

In $\mathbb{R}^4$ let us consider the vector subspace

$$V = \{ (x, y, z, w) \mid x + y + z - w = 0, x + 3y = 0 \}.$$

Which one of the following statements is true?

- ☒ (a) V is the vector subspace generated by the vectors $(1, 1, 1, -1)$ e $(1, 3, 0, 0)$. ✖
- ☐ (b) V is the vector subspace generated by the vectors $(3, -1, 0, 2)$ e $(-3, 1, 2, 0)$.
- ☐ (c) V is a line of $\mathbb{R}^4$.
- ☐ (d) V is a subspace of dimension 1.

Risposta errata.

The correct answer is: V is the vector subspace generated by the vectors $(3, -1, 0, 2)$ e $(-3, 1, 2, 0)$.

Question 7

Incorrect

Mark -15.00 out of
100.00

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the endomorphism $f(x, y) = (-x + 2y, -y)$.

Which one of the following statements is true?

- ☒ (a) f is simple. ✖
- ☐ (b) f has two distinct eigenvalues.
- ☐ (c) $(0, 1)$ is an eigenvector.
- ☐ (d) $(2, 0)$ is an eigenvector.

Risposta errata.

The correct answer is: $(2, 0)$ is an eigenvector.

Question 8

Incorrect

Mark -15.00 out of
100.00

In the euclidean 2-dimensional space with a fixed cartesian system, consider the conic $\mathcal{C}$ having equation

$$x^2 - 6xy + 9y^2 + 2x - 4y + 1 = 0.$$

Which of the following statements is true?

-
- ☒ (a) $\mathcal{C}$ is degenerate. ✖
- ☐ (b) $\mathcal{C}$ is a parabola.
- ☐ (c) $\mathcal{C}$ is a hyperbola.
- ☐ (d) $\mathcal{C}$ is an ellipse.

Risposta errata.

The correct answer is: $\mathcal{C}$ is a parabola.

Question 9

Correct

Mark 100.00 out of
100.00

In the euclidean 3-dimensional space with a fixed cartesian system, let us consider the sphere $\mathcal{S}$ and the plane π of equations respectively

$$\mathcal{S} : x^2 + y^2 + z^2 - 4z = 0, \quad \pi : x - z + 2 = 0.$$

Which of the following statements is true?

-
- ☒ (a) The radius of the circumference $\mathcal{S} \cap \pi$ is equal to 2. ✔
- ☐ (b) $(0, 2, 0) \in \mathcal{S} \cap \pi$.
- ☐ (c) π is tangent to $\mathcal{S}$.
- ☐ (d) The distance of the center of $\mathcal{S}$ from the plane π is equal to 1.

Risposta corretta.

The correct answer is: The radius of the circumference $\mathcal{S} \cap \pi$ is equal to 2.

Question 10

Correct

Mark 100.00 out of
100.00

Let us consider the quadratic form defined by

$$q(x, y) = (x, y)A \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{with} \quad A = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}.$$

Which of the following statements is true?

-
- ☒ (a) $q(a, b) \geq 0$ for every $(a, b) \in \mathbb{R}^2$. ✓
 - ☐ (b) The columns of A are eigenvectors for A .
 - ☐ (c) The matrix A has two negative eigenvalues.
 - ☐ (d) There exists $(a, b) \in \mathbb{R}^2$ such that $q(a, b) < 0$.

Risposta corretta.

The correct answer is: $q(a, b) \geq 0$ for every $(a, b) \in \mathbb{R}^2$.